

USBasp to BTE13-010A Pro Mini

Six wires. Every pin you need is on the board's top row. Nothing else connected: no FTDI, no battery, no string.

1 The USBasp 10-pin header, looking at the connector

odd row: 1 3 5 7 9

1 MOSI	3 NC	5 RST	7 SCK	9 MISO
2 VCC	4 GND	6 GND	8 GND	10 GND

even row: 2 4 6 8 10

Pin 1 is marked with a dot, a triangle or a square pad. On a ribbon cable it is the red stripe. Only 1, 2, 5, 7, 9 and one GND are used.

2 The Pro Mini top row, board the right way up, FTDI header on the left

RAW	GND	RST	VCC	A3	A2	A1	A0	13	12	11	10
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Coloured pins are the six you use. A4 to A7 are the unpopulated holes at the far end; ignore them.

3 The six connections

USBasp	Signal	Pro Mini pin	Suggested wire
Pin 1	MOSI	11	green
Pin 9	MISO	12	blue
Pin 7	SCK	13	yellow
Pin 5	RESET	RST	orange
Pin 2	VCC	VCC	red
Pin 4, 6, 8 or 10	GND	GND	black

RESET goes to RST, not to a numbered pin. MOSI and MISO are the pair people swap: 11 out, 12 back.

4 USBasp jumpers, then the commands

JP1 or a "3V3/5V" pad pair: leave at 5 V. This board is a 16 MHz Pro Mini, so it programs at 5 V.
JP2 is only for updating the USBasp's own firmware. Leave it open.
JP3, "slow SCK": fit it if the chip will not answer. Old clone firmware ignores avrdude's -B option.
Plug the USBasp into the Mac. No driver is needed on macOS. Test first, this writes nothing:

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avrdude -c usbasp -p m328p -v
Expect: Device signature = 0x1e950f (probably m328p)
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Write the fuses, then read them back:
avrdude -c usbasp -p m328p -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m
avrdude -c usbasp -p m328p -U lfuse:r:-:h -U hfuse:r:-:h -U efuse:r:-:h
Expect 0xe2, 0xd9, 0xfd. The last one often reads back as 0x05: only three bits exist, same value.
avrdude lives with the Arduino IDE at ~/Library/Arduino15/packages/arduino/tools/avrdude/*/bin/,
or install your own with: brew install avrdude

5 If it will not talk

Message	Usual cause
could not find USB device 0x16c0/0x5dc	USBasp not enumerated: cable, hub, or a dead clone
device signature = 0x000000	No power at the target, or a loose wire. Check VCC on the board
signature reads as 0xfffff	MISO not connected, or MOSI and MISO swapped
initialization failed, rc=-1	SCK too fast: fit JP3, or try -B 8
warning: cannot set sck period	Harmless on old clone firmware. Use JP3 instead

Never program RSTDISBL, and never select an external clock you do not have. Both lock the chip out of ISP.